Tracker Reminder - Requirements

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1. Purpose and scope

Purpose: automatically email people the drawings assigned to them, on the day each is due, using any project tracker Excel file stored on SharePoint.

In scope

- Read tracker workbooks from SharePoint and find assignments per role: **Engineers, Modelers, Annotation, QA/QC**.
- Send each assignee one email per run listing the drawings due today.
- Send one overdue reminder per item.
- Send the lead a nightly summary.
- Provide an admin web page to register trackers, tune name matching and preview emails.

Out of scope (for now)

- Editing the tracker Excel files.
- Reminders on channels other than email.
- Excel add-in (planned later phase, see section 6).

Reference behaviour: the logic must match the existing Excel_Notify add-in exactly, reproduced in JavaScript.

2. Users and roles

Role	Who	What they do
Assignee	Engineers, Modelers, Annotation, QA/QC staff	Receive due-today and overdue emails. No login needed.
Lead	Per tracker	Receives the nightly summary.
Admin	Listed in `admins.json` (first: Vasanthan, mirage.automation)	Use the admin page: trackers, name matching, preview, admin list.

Role	Who	What they do
IT	Mirage AEC IT	Owns the Entra app, shared mailbox, Exchange scope, SharePoint permissions.

3. Functional requirements

3.1 Reading trackers

- **FR-1** Read a tracker `.xlsx` from SharePoint via Microsoft Graph (address-bar, library and sharing links).
- **FR-2** Detect each sheet's layout (header row, columns) as Excel_Notify does; the layout is not fixed.
- **FR-3** Support four roles per sheet: Engineers, Modelers, Annotation, QA/QC, each with its own name and date columns.
- **FR-4** Parse multi-name cells and date cells exactly as Excel_Notify (same rules, ported 1:1).

3.2 Name matching

- **FR-5** Resolve a name to an email in this order: shared **name map** in SharePoint, then the tracker's **People sheet**, then the **Microsoft 365 directory**.
- **FR-6** Only `@mirageaec.com` addresses are allowed.
- **FR-7** Unresolved names are reported to the lead and shown in Preview; they are never guessed.

3.3 Reminder rules

- **FR-8** On an item's due date, email each assignee their drawings due today.
- **FR-9** Each overdue item gets **one** reminder only.
- **FR-10** Items already overdue when a tracker is registered follow a per-tracker policy chosen in the UI: `afterRegistration`, `emailOnce` or `leadSummaryOnly`.
- **FR-11** No duplicate sends: a sent-log keyed by tracker + item + type (due/overdue) + email.
- **FR-12** The lead gets a nightly summary per tracker.

3.4 Email sending

- **FR-13** Send from the shared mailbox `project-reminders@mirageaec.com` by default ("on behalf of" mode).
- **FR-14** Later: optional "send as me" via the user's own sign-in.
- **FR-15** Email carries the Mirage logo and an HTML table of drawings.

3.5 Scheduling

- **FR-16** Each tracker has its own send time in **IST**, 30-minute steps, default **8:00 PM**, editable on the page.
- **FR-17** The job wakes every 30 minutes and runs only trackers that are due and have not run today.
- **FR-18** Last-run state is stored in `runstate.json`.

3.6 Admin web page

- **FR-19** Screens: **Trackers**, **Name matching**, **Preview**, **Admins**.
- **FR-20** Sign in with Microsoft; only listed admins get in.
- **FR-21** Saves use eTag so two admins cannot overwrite each other.
- **FR-22** Preview shows exactly who would get which email, without sending.

3.7 Test mode

- **FR-23** A global switch **or** a per-tracker switch redirects every email to a test address with the subject prefix `[TEST -> x]`.
- **FR-24** Test sends use their own sent-log keys, so they never block real sends.

3.8 Alerts

- **FR-25** Problems (bad config, unreadable tracker, sign-in failure) email the alert address, at most once per day per problem.

4. Non-functional requirements

- **NFR-1 Least privilege:** the app can send only as `project-reminders@` (Exchange scope `TrackerReminderSender`); SharePoint access via `Sites.Selected` (write on Automation, read on the BIM site).

- **NFR-2 No secrets in code:** the client secret lives in GitHub Secrets / local `.env` only, never in the repo or the page.
- **NFR-3 Parity:** output must match Excel_Notify for the same workbook (checked by `tools/parityCheck.js`).
- **NFR-4 Reliability:** Graph calls time out at 60 s and retry on 429/503/504; the sent-log prevents duplicates after a rerun.
- **NFR-5 Auditability:** sent-log and run-state are kept in SharePoint; each run prints step-by-step progress.
- **NFR-6 Maintainability:** SOLID design, simple readable code, a CHANGE LOG comment in every file (why, what, bug and cause).
- **NFR-7 Safe default:** test mode is ON until both switches are turned off deliberately.
- **NFR-8 Portability:** core logic is plain ES modules shared by the Node job and the browser page, so moving hosts (GitHub to netcup) changes only the scheduler.

5. Constraints and assumptions

- **Tenant:** Microsoft 365 (mirageaec.com); SharePoint sites `/sites/Automation` and `/sites/MirageAEC_Engg_BIM`.
- **Hosting (test):** GitHub Actions (job) and GitHub Pages (page). Target later: netcup.
- **GitHub cron is best-effort:** runs can be late or dropped; the schedule uses :07/:37 to reduce this.
- **Time:** all send times are IST; dates are handled as ISO `YYYY-MM-DD` strings.
- **Assumption:** trackers follow the Excel_Notify layout rules (four roles, detectable header row).
- **Assumption:** assignee names can be resolved to `@mirageaec.com` mailboxes.
- **Constraint:** the page reads Excel in the browser (SheetJS), so very large workbooks may be slow.

6. Status, open items and later phases

Done and proven

- Core logic ported from Excel_Notify; parity check passes.
- End-to-end send via the shared mailbox, test redirection, no duplicates.
- GitHub job runs green; admin page published on GitHub Pages.

Open

- [ ] Confirm GitHub fires the **scheduled** runs (3:00 PM test on 2026-10-14).
- [ ] Set the test tracker's send time back to 8:00 PM.
- [ ] Click-test Name matching, Preview, Add tracker and sharing-link resolution.
- [ ] Shorten the test-mode label on the Trackers page.
- [ ] Enable GitHub 2FA before 2026-10-18.
- [ ] Register the live BIM tracker; turn both test switches off.

Later phases

- **Excel add-in:** separate installer, same "Mirage AEC" ribbon tab, WebView2 side panel plus "Open in browser", prefilled with the workbook's SharePoint link.
- **"Send as me"** sender mode.
- **Move to netcup** if GitHub scheduling proves unreliable.